

Linear Circuit Analysis

(EE1001)

Date: April 3rd 2024

Course Instructor(s)

- | | |
|------------------------|------------------|
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Sessional-II Exam

Total Time (Hrs):	1
Total Marks:	40
Total Questions:	2

Roll No

Section

Student Signature

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1. Attempt all the questions.
2. Attempt all parts of the same question together.
3. Show all the steps with proper circuit diagrams and answers with proper units.

CLO # 02: Apply DC circuit analysis techniques and theorems to solve linear resistive circuits.

- Q1:** For the circuit shown in Figure 1 below, [20 marks]
- (a) Calculate the Thévenin Equivalent voltage, V_{Th} with respect to the terminals a,b. [8]
 - (b) Calculate the Thévenin Equivalent resistance, R_{Th} at a,b. [8]
 - (c) Sketch the Thévenin and Norton Equivalent circuits. [4]

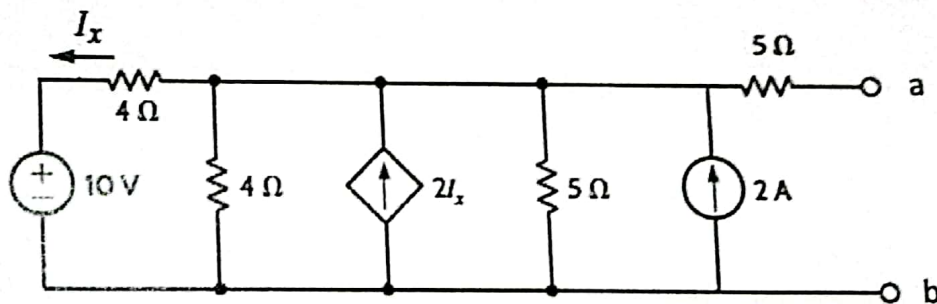
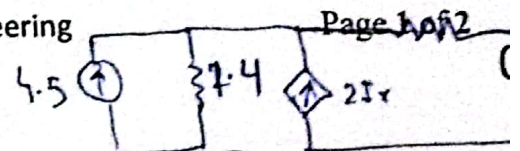


Figure 1. Circuit for Thévenin Equivalent

$$I_x = \frac{V_{Th}}{4}$$

$$\begin{aligned} -i_1 + i_n + i_2 &= 0 \\ -i_1 + i_n + i_2 &= 0 \\ \boxed{i_2 = i_1} \end{aligned}$$



Spring 2024

Department of Electrical Engineering

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CLO # 03: Solve circuits containing ideal operational amplifiers

- Q2:** For the circuit shown in the Figure 2 below, the op amp is ideal. **[20 marks]**
- (a) Label the five op amp terminals with their names. (Draw a separate diagram for the circuit symbol and then label it). **[5]**
- (b) Express the output voltage v_o in terms of the input voltage sources; v_a and v_b by assuming linear region of operation. **[10]**
- (c) If $v_b = 2\text{ V}$, determine the range of v_a such that the amplifier does not saturate. **[5]**

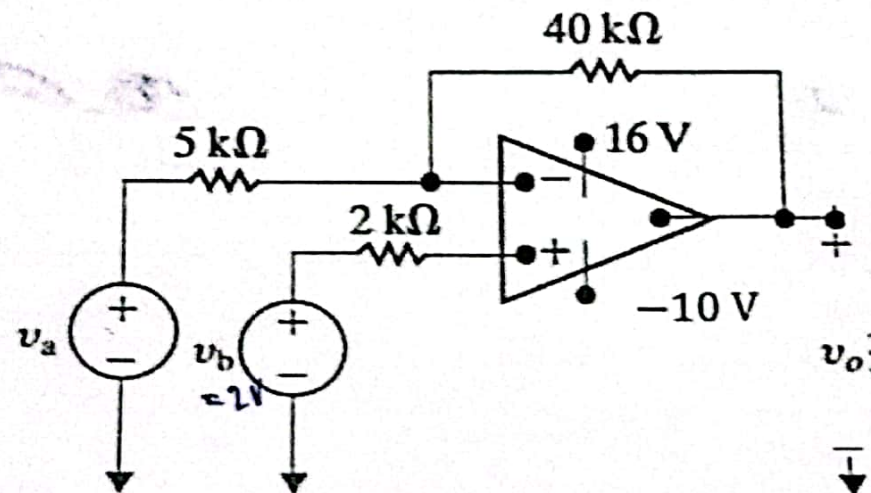


Figure 2. An Operational Amplifier Circuit